

Paper Manager pack — GR00T N1 (Oct 7)

Paper: GR00T N1: An Open Foundation Model for Generalist Humanoid Robots

arXiv: <https://arxiv.org/abs/2503.14734>

Short: GR00T N1 / GR00T-N1-2B (~2.2B)

PDF: /workspace/papers/gr00t-n1-2503.14734.pdf

Session: Oct 7 · due 9:30 AM · **all seats empty** — full five packs for club/GitHub

Sources: Summarizer + Senior Research (/workspace/groot-n1-quiet/DELIVERABLE.md)

1. Paper summary

Dual-system humanoid VLA: **Eagle-2** vision-language **System 2** (~10 Hz) + flow-matching **DiT System 1** (**120 Hz** actions) via **cross-attn**, jointly trained. Chunk **H=16**, Euler **K=4**; **~63.9 ms**/chunk on L40. Heterogeneous data pyramid (human/latent + DexMimicGen + neural I2V + real GR-1/OXE/AgiBot). Beats BC-Transformer & Diffusion Policy in sim/real; strong low-data efficiency. Deployed on Fourier **GR-1**.

Lineage (CLEAR): N1 → Isaac → N1.6-BEHAVIOR1k → **N1.7 GA** = official **BEHAVIOR'26 co-baseline with $\pi_{0.5}$** . Do **not** conflate N1 paper numbers with N1.7 challenge baseline (APIs/horizons differ).

2. Keywords

GR00T N1, humanoid VLA, dual-system, Eagle-2, FM DiT, data pyramid, N1.7, BEHAVIOR 2026, π_0 / $\pi_{0.5}$ contrast, LAPA/IDM, DexMimicGen

3. Architecture vs $\pi_0/\pi_{0.5}$

Axis	GR00T N1	$\pi_0/\pi_{0.5}$
Coupling	Separate DiT **cross-attn**	Action expert **shared self-attn** + block causal
VLM	Eagle-2 (SmolLM2+SigLIP-2)	PaliGemma

Chunk	H=16, K=4; 10/120 Hz	H=50 (π_0), ~10 Euler; 20–50 Hz
I/O	Per-embodiment MLPs; LAPA/IDM	Pad/mask shared dim
Successor	**N1.7** ~3B, H≤40, Apache-2.0	$\pi_{0.5}$ + RTC + BEHAVIOR’25 forks

4. Role packs

Stakeholder (18+3)

Open humanoid VLA path → Apache N1.7 + BEHAVIOR’26 parity with $\pi_{0.5}$. Prefer finetune nvidia/GR00T-N1.7-3B. Risks: N1≠N1.7 API; short-horizon paper vs house-scale; forgetting; Isaac GPU needs.

Scientific Reviewer (8+3)

Strengths: falsifiable dual-system thesis; data methodology; broad eval; honest forgetting fail; open formats.

Weaknesses: Table2/4 DexMG mismatch (preserve both numbers); IL not VLA baselines in main; small-n partial credit; confounded ablations; short-horizon rhetoric vs “generalist humanoid.”

Verdict: Strong systems paper; architectural foil to π MoE coupling.

Empiricist (11+3)

H1: K=4 near-optimal; flatten by ~4–8

Plus: adapter low-data lift vs scratch DP/BC (directional)

Priority	Experiment
P0 Y	Released N1-2B dry infer; eval-only NFE/chunk grid ($K \in \{1,2,4,8\}$, H)
P0 Partial	Mid-layer vs final VL features
P1 Partial	Adapter-only FT (30–100 demos); data-regime curve
Defer N	Full pretrain (~50k H100-h), full RoboCasa×GR-1, real GR-1, BEHAVIOR’26 N1.7 from scratch

Tattoo: sim avg ~45% vs DP 33.4% / BC-T 26.4%; real 76.8% vs DP 46.4%. Env: /workspace/groot-n1-quiet/env_outline/.

Archaeologist (11+3)

Priors: SigLIP-2, SmolLM2, Eagle-2, Lipman FM, π_0 (cite+contrast), Diffusion Policy, ACT, DiT/Flamingo/VIMA, LAPA, IDM, DexMimicGen/OXE.

Pivot: humanoid FM-VLA + dual-system cross-attn + data pyramid.

Genealogy: SigLIP-2+SmolLM2 → Eagle-2 → **N1** → N1.6 → **N1.7** → BEHAVIOR'26 $\parallel \pi_{0.5}$.

Visionary (11+3) — HIGH for BEHAVIOR'26

1. **N1.7 is official substrate** — dual-track vs $\pi_{0.5}$ from day one
2. Explicit System-2 interface → hierarchy/safety hooks vs fused π expert
3. Port neural pyramid ideas + RTC/correlated-noise hybrids from π forks
4. Version discipline: paper H=16 vs N1.7 $H \leq 40$ — don't mix APIs
5. Try both starters on hard BEHAVIOR tasks; report NFE/horizon; with/without System-2 mid-layer
6. **Durable idea:** modular VL ↔ motor coupling + open humanoid data pyramid — not just a bigger PaliGemma clone

Skip Discord/Drive.

5. Control note

Dual System 2 (VLM mid features) + System 1 (DiT action CFM) — **not** a classical planner. Contrast OpenPI action-expert FM, DP CNN/Transformer, π _RL online PPO, Comet RFT for BEHAVIOR.

6. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	N1 → N1.7 product path
Reviewer	8+3	Dual-system thesis + Table mismatch
Empiricist	11+3	K/H smoke grid + tattoo numbers
Archaeologist	11+3	Genealogy to N1.7 / BEHAVIOR'26
Visionary	**11+3**	**Dual baseline with $\pi_{0.5}$ (HIGH)**

Spine: dual-system cross-attn FM; N1≠N1.7. PDF for Mac.

GitHub title: [ECE 605] - GR00T N1 Making Club Pack

Paths

Artifact	Path
This pack	`/workspace/groot-n1-quiet/KANTA_PACK.md`
Senior Research	`/workspace/groot-n1-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/gr00t-n1-2503.14734-summary.md`
Paper PDF	`/workspace/papers/gr00t-n1-2503.14734.pdf`